

## COMPLETE SYSTEM MAP

# ENRK

An over-collateralised stablecoin whose unit of account is one kilowatt hour of thermodynamic value — not a fiat currency.

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### Unit of account

1 ENRK = 1 kWh

### Collateral

KAS, over-collateralised

### Governance

None. Frozen at compile time.

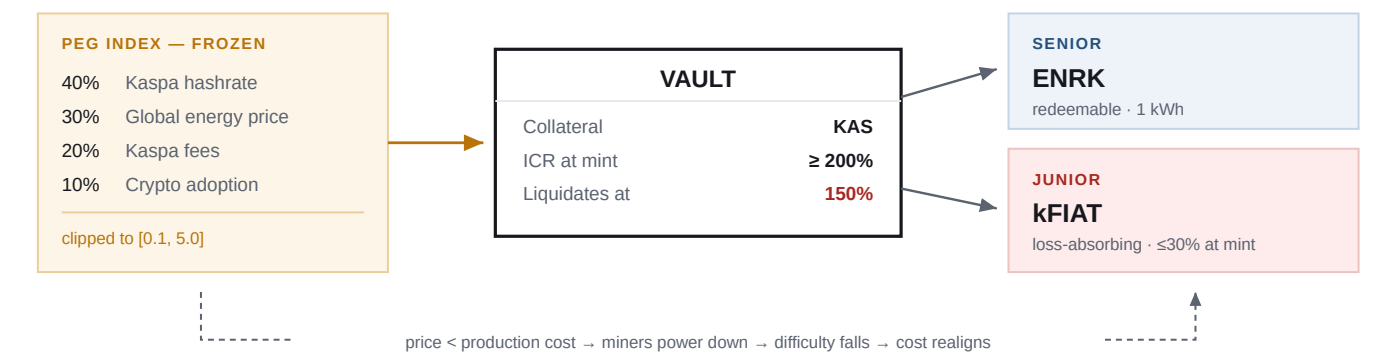
### Execution layer

Kaspa L1 — pending validation

**Status:** design and specification complete, implementation not started. No code is deployed, no audit has been performed. The covenant oracle construction is derived from KIP text and has **not** been validated by the authors of those KIPs. Nothing here is an invitation to use the protocol.

# 01 The protocol in one page

Collateral and unit of account are causally linked through proof-of-work. That link is what makes Kaspa the right chain rather than an arbitrary one.



**Figure 1.** The peg index feeds vault pricing; vaults issue two tranches against KAS collateral. The dashed path is the thermodynamic feedback loop that no proof-of-stake chain can offer.

## Frozen parameters

| PARAMETER               | VALUE            | PARAMETER        | VALUE                 |
|-------------------------|------------------|------------------|-----------------------|
| ICR minimum at mint     | 200%             | Mint fee         | 2%                    |
| MCR liquidation trigger | 150%             | Liquidation fee  | 4%                    |
| kFIAT cap at mint       | 30% of debt      | Redemption fee   | 1%                    |
| Auction duration        | 120 min          | Fee denomination | KAS, from collateral  |
| Auction start → floor   | 100% → 75%       | Treasury share   | 20% → 0% at cap       |
| Redemption ordering     | lowest ICR first | Treasury cap     | 2,500,000 peg units   |
| Redemption min ICR      | 100%             | Circuit breaker  | 10% dev / 6h downtime |

**THE ONE GOVERNANCE MOMENT**

There is exactly one decision anyone gets to make: deploy, or do not deploy. After that the rules are law. Disagreement is resolved by forking to a v2 that competes on merit, not by voting to change v1 under whoever holds the most tokens.



## 03 Who uses it, and why they bother

A protocol with no governance has no way to make anyone do anything. Every function that must happen has to be somebody's most profitable available action. Where that is not true, the mechanism does not work — section 08 is the case where it stops being true.

### DEMAND SIDE

#### Households in inflating economies

Hold ENRK instead of a currency losing double digits a year. They get a unit nobody can dilute and no issuer can freeze.

**Honest limits:** ENRK is volatile against their grocery bill; it needs local on/off ramps that may not exist; and in a KAS crash the senior tranche can still take a loss. It replaces one risk with a different, smaller, more visible one — not with safety.

### SUPPLY SIDE

#### Miners

Hold the asset, know their real cost per kWh, and are the natural arbitrageurs: when ENRK trades above peg, minting against KAS and selling is directly profitable; when it trades below, buying and redeeming at par is.

**Why it self-corrects:** their arbitrage adds supply exactly when ENRK is rich and removes it exactly when ENRK is cheap. Nobody pays them to defend the peg — defending it is the trade.

### CLEARING

#### Liquidators

Buy collateral from failing vaults at a discount via the Dutch auction. Modelled demand is  $4\% + 18\% \times \text{drawdown}$  — the deeper the crash, the more discount they require to act.

**The binding constraint:** their capital, not their willingness. Modelled at 5% of debt clearable per day. This is what actually broke on Black Thursday, and it is what breaks here.

### PEG DEFENCE

#### Redeemers

Burn ENRK for KAS at peg, minus 1%. Profitable whenever ENRK trades below peg, which is exactly when the peg needs defending.

**Why it is the strongest mechanism:** it needs no external capital — the redeemer already holds the ENRK. It works when auctions cannot. It is also the one mechanism that has never been implemented.

### The miner arbitrage loop

#### ENRK ABOVE PEG



#### ENRK BELOW PEG



**Figure 3.** Both loops are voluntary and both are profitable. This is the Terra comparison that matters: Terra's arbitrage was backed by a token whose only value was the arbitrage itself. Here the collateral is KAS, whose production cost exists whether or not ENRK does.

### WHY THIS IS NOT TERRA

Terra's LUNA had no value independent of UST's demand for it — a circle. ENRK's collateral is an asset with an exogenous production cost set by electricity and difficulty. When the circle broke at Terra, there was nothing underneath. Here, what is underneath is a mining industry with a marginal cost.

# 04 Two tranches, and who eats the loss

One vault issues two claims against the same collateral. ENRK is senior and redeemable. kFIAT is junior, absorbs losses first, and is deliberately not redeemable. This is the structure that lets ENRK stay whole while KAS falls.

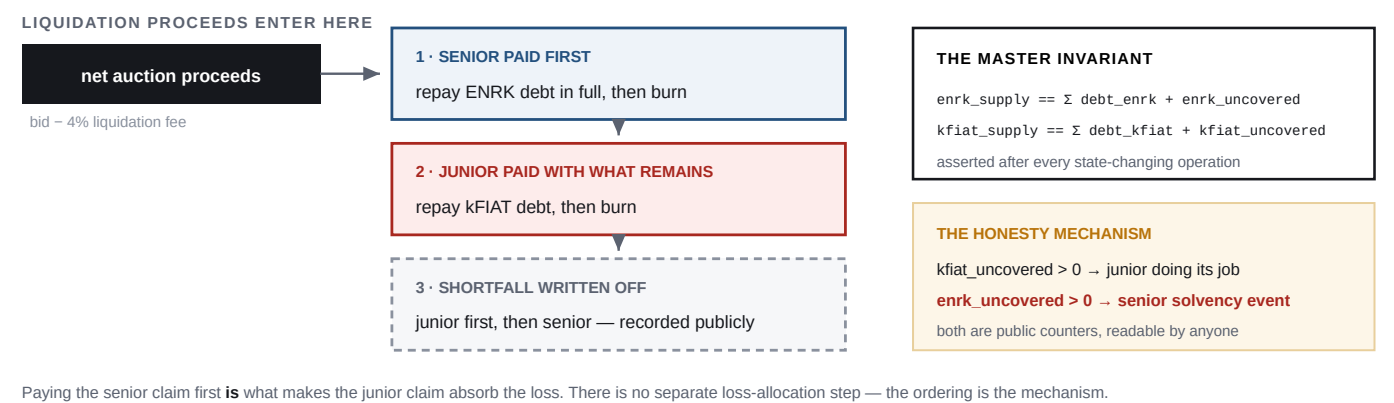


Figure 4. The seniority waterfall. Write-offs are published rather than absorbed silently: a protocol that cannot be repaired must at minimum be legible.

## kFIAT — the three things that must be said plainly

NOT A GUARANTEE

The 30% cap is a mint-time ceiling

It is a *ratio*, so it can be breached without minting anything — by shrinking the denominator. Burning ENRK does exactly that, and ENRK burns are core paths: repayment, redemption, buyback. Peg defence and the cap pull against each other, and peg defence must win. Public documentation must say "**≤ 30% at mint**", never "always ≤ 30%".

DELIBERATE ASYMMETRY

kFIAT cannot be redeemed

In a crisis, ENRK holders can exit at par and kFIAT holders cannot. Stress tests show kFIAT benefits *indirectly* — redemption stops vaults from reaching shortfall at all — but in a deeper scenario ENRK drains the collateral and kFIAT faces the remainder. That is seniority working as designed. It must be documented as a property, not discovered by a holder.

WHO BUYS IT

The junior tranche is a leveraged bet

kFIAT holders take first loss in exchange for leveraged exposure to the energy thesis. They are speculators, and the design says so. Their willingness to be first in line is what makes the senior tranche credible — the same trade as any subordinated debt.

## Where the buffer holds, and where it does not

| SCENARIO         | LATENT HOLE | KFIAT BUFFER | OUTCOME FOR ENRK                                    |
|------------------|-------------|--------------|-----------------------------------------------------|
| KAS -85%, median | 16%         | 30%          | Comfortably covered                                 |
| KAS -85%, p95    | 45%         | 30%          | Buffer exceeded — ENRK structurally exposed to ~21% |
| KAS -95%, median | 30%         | 30%          | At the limit                                        |
| KAS -95%, p95    | 57%         | 30%          | Buffer exceeded by a wide margin                    |

Figures from the pre-fix model (85% auction floor, no redemption). Section 08 shows what the corrected parameters do to them.

## 05 The five equilibrium mechanisms

Five forces are supposed to hold the peg. Three exist, one is broken, one was never written. Stating which is which is the point of this page.

| # | MECHANISM                                             | ACTS WHEN                 | NEEDS EXTERNAL CAPITAL       | STATE                       |
|---|-------------------------------------------------------|---------------------------|------------------------------|-----------------------------|
| 1 | <b>Redemption</b> — burn ENRK, receive KAS at peg -1% | ENRK below peg            | No                           | <b>NEVER BUILT</b>          |
| 2 | <b>Miner arbitrage</b> — mint/sell or buy/redeem      | Either direction          | Their own                    | <b>EXTERNAL, WORKS</b>      |
| 3 | <b>Stability Pool buyback</b>                         | Panic                     | Yes                          | <b>BUILT, DEFECTIVE</b>     |
| 4 | <b>Dutch auction</b> — permissionless liquidation     | ICR < 150%                | Yes — the binding constraint | <b>BUILT, TESTED, FIXED</b> |
| 5 | <b>PoW difficulty adjustment</b>                      | KAS below production cost | No                           | <b>EXTERNAL, AUTOMATIC</b>  |

### KNOWN DEFECT, CARRIED FORWARD

#### The Stability Pool cannot buy

attempt\_buyback computes a cost, never debits it, and reduces the pool's own ENRK balance. It is a burn, not a bid: no order is placed, no sell pressure absorbed.

Worse, its ammunition was denominated in the asset it defends — its purchasing power collapses with the crisis it exists to fix. That is the same error, structurally, that this design rejects everywhere else. **Fix it or remove it. For immutable code, removing is safer.**

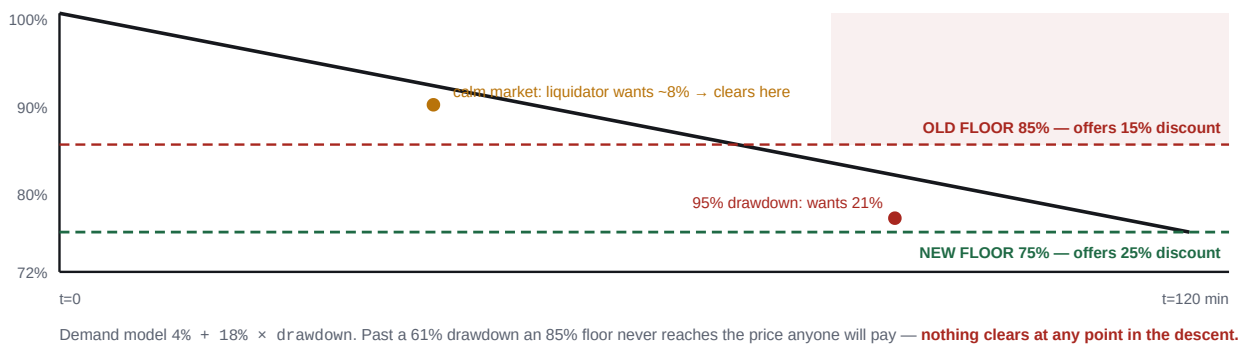
### THE MECHANISM THAT REPLACES IT

#### Redemption supersedes the pool

Anyone converting ENRK to KAS at par is a stronger floor than a pool buying at a discount — and the redeemer brings their own capital, so it works precisely when liquidators have run out.

**Its structural limit:** at face value new ratio =  $(a-r)/(b-r)$  improves a healthy vault and degrades an underwater one. Below 100% ICR the collateral to honour face value does not exist. At -95%, 26% of redemption demand finds no eligible vault — the wall arrives exactly when the mechanism is most needed.

## Dutch auction — why the floor is the whole game

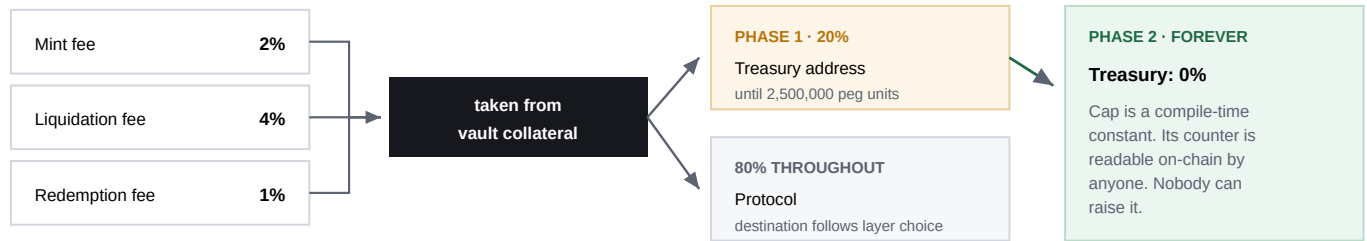


**Figure 5.** A Dutch auction clears at the first acceptable price, not at the floor — so floors of 75%, 70% and 60% give *identical* results. The floor is a safety valve, not a price. 75% is the shallowest one that always clears, and therefore the correct choice.

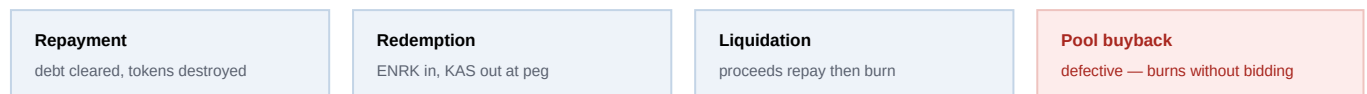
## 06 Fees, burns, and the treasury that ends

Every fee is taken in KAS out of vault collateral, never minted as ENRK. That single choice keeps the master invariant exact, adds no supply inflation, and makes the protocol accumulate an asset it does not itself issue.

### SOURCES — ALL DENOMINATED IN KAS



### SEPARATELY — WHERE ENRK IS BURNED



**Figure 6.** Fee flow and burn paths. Note the asymmetry: fees accrue in KAS, an asset the protocol does not issue; burns destroy ENRK, the asset it does. Nothing about a fee changes total ENRK supply.

#### Why KAS and not minted ENRK

1. The master invariant becomes exact:  $\text{enrk\_supply} == \sum \text{debt\_enrk}$ . No fee-minting, no rounding remainder splitting 80/20 into 2 units out of 3.
2. No supply inflation from fees, ever.
3. The protocol accumulates an **exogenous** asset — the lesson the Stability Pool taught the hard way.

Cost: taking KAS lowers the vault's ICR, so the preflight check must account for it. An implementation detail, not a design problem.

#### Why the treasury is capped by amount, not by time

A time-based taper is a bet on adoption speed — too slow and it raises nothing, too fast and it raises ten times what is needed. An amount cap is self-limiting and publicly verifiable, and the only way to "game" it is to reach it faster, which ends the treasury sooner at the gamer's expense.

An immutable protocol has nothing to fund after launch: it ships once and can never be updated. Costs are front-loaded. **A perpetual claim with no corresponding obligation is a rent.**

#### DISCLOSED, NOT DISCOVERED

The 2.5M figure (~\$300k at ~\$0.12/kWh) is justified line by line: economic audit 30–70k, legal review 10–30k, five years infrastructure 15–40k, frontend 20–50k, bug bounty 50–100k, development 80–120k — assuming code audits are ecosystem-supported. If they are not, there is a shortfall to be found elsewhere. This is stated here rather than found later, because an undisclosed founder revenue is what destroys trust — not taking one.

## 07 Attack surface, without exception

Every vector considered, including the ones with no clean answer. An immutable protocol cannot patch its way out of anything on this page, so the list is written to be complete rather than reassuring.

| VECTOR                               | MECHANISM                                                                                                                    | DEFENCE                                                                                                                                                                                          | RESIDUAL                       |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <b>Stale price</b>                   | Liquidator picks an old favourable price and liquidates vaults that are healthy at the true price. Theft, not inconvenience. | Sweep and republish occur in <b>one transaction</b> . Round R is consumed and R+1 created atomically — there is no instant where both exist.                                                     | <b>CLOSED</b>                  |
| <b>Oracle feed forgery</b>           | Attacker publishes a fake price UTXO.                                                                                        | Authenticity by covenant lineage: <code>OpInputCovenantId(1) == ORACLE_COVENANT_ID</code> . Not a signature check — Kasper script has no opcode to verify a signature over an arbitrary message. | <b>UNVALIDATED</b>             |
| <b>Oracle DoS / feed drain</b>       | Attacker consumes every price UTXO so nobody can liquidate.                                                                  | The USE branch forces identical replication — a spender must recreate what they consume. Condition (a) additionally forces the attacker to touch a real vault.                                   | <b>CLOSED</b>                  |
| <b>Oracle price manipulation</b>     | Corrupt the underlying indices to mint cheaply or force liquidations.                                                        | Two of four indices are chain-native and unforgeable (hashrate, fees). Circuit breaker at 10% peg deviation, 6h feed downtime — no override, no multi-sig.                                       | <b>BOUNDED</b>                 |
| <b>Governance capture</b>            | Whales vote parameters in their own favour — Terra, MakerDAO collateral policy, Curve via Yearn/Convex.                      | There is no governance to capture. Every parameter is a compile-time constant.                                                                                                                   | <b>STRUCTURALLY IMPOSSIBLE</b> |
| <b>Admin key compromise</b>          | Attacker obtains the upgrade key and drains or rewrites.                                                                     | No admin key exists. No upgrade path exists. An attacker cannot use what was never built.                                                                                                        | <b>STRUCTURALLY IMPOSSIBLE</b> |
| <b>Custodial bridge seizure</b>      | Bridge signers collude and empty the vault holding all collateral.                                                           | Native L1 — the collateral never moves off Kasper. This is the entire reason the L1 path was set aside.                                                                                          | <b>AVOIDED BY DESIGN</b>       |
| <b>Reflexive liquidation spiral</b>  | Liquidated collateral is dumped, pushing KAS lower, triggering more liquidations.                                            | Modelled explicitly with a square-root impact law. Dutch auction spreads selling over 120 minutes rather than dumping at market.                                                                 | <b>REAL, MODELLED</b>          |
| <b>Liquidator capital exhaustion</b> | Nobody has the capital to bid; auctions fail; debt goes uncovered.                                                           | 75% auction floor covers maximum modelled demand. Redemption needs no liquidator capital at all.                                                                                                 | <b>THE CORE RISK</b>           |
| <b>Bank-run on a reserve</b>         | A finite, visible reserve paying first-come-first-served makes exiting early the rational move.                              | No such reserve exists. The insurance fund was analysed and <b>rejected</b> for exactly this — see section 09.                                                                                   | <b>REJECTED BY DESIGN</b>      |
| <b>51% attack on Kasper</b>          | Hostile actor rewrites blocks.                                                                                               | None specific to ENRK. This is a Kasper-wide failure, and ENRK is not a solution to it any more than an ERC-20 is a solution to an Ethereum reorg.                                               | <b>INHERITED</b>               |
| <b>Energy price shock</b>            | Energy index spikes during a crash, holding the peg firm while collateral collapses — maximising the fall in ICR.            | None. This is the MakerDAO Black Thursday dynamic, and it is the documented cost of the 30% uncorrelated hedge.                                                                                  | <b>ACCEPTED, PRICED</b>        |
| <b>Immutable bug</b>                 | A critical defect ships and can never be patched.                                                                            | No defence exists. Only pre-deployment scrutiny and, afterwards, migration to a v2.                                                                                                              | <b>UNMITIGABLE</b>             |

### THE SUBSIDY THAT WAS ALMOST BUILT

A rule of the form "top up any liquidation shortfall from the reserve" converts oracle errors from *someone loses* into *the reserve pays* — which **subsidises attacks on the oracle**. Adding a mechanism whose failure dynamic is already known is worse, in immutable code, than adding nothing.

### WORST CASE, STATED ONCE

The oracle's worst case is mispriced liquidations: bad, bounded, recoverable. A custodial bridge's worst case is total loss of collateral. Those are different *categories* of risk, and that difference is the whole argument for the L1 path despite it being first-of-its-kind.

## 08 What we found wrong with our own design

A Monte Carlo stress test of the whole protocol — 1000 vaults, realistic bidder scarcity, the seniority waterfall, reflexive price impact, gap risk, and a liquidator capital constraint. It changed the design. It is published here rather than discovered later.

### THE MODEL WAS WRONG THREE TIMES BEFORE IT WAS RIGHT

1. A failed auction destroyed the collateral — it destroys nothing; the vault stays open and is re-auctioned. 2. Linear market impact turned a -50% exogenous crash into -85% actual, flattening every comparison — replaced with a square-root law. 3. Liquidator capital was unlimited, so the system always cleared — added a daily capacity constraint, which is what actually broke on Black Thursday. **Before those fixes it reported "everything is fine" for entirely wrong reasons.**

### The central finding: it freezes, it does not explode

|                                                                  |                                                                                |                                                                       |                                                                         |
|------------------------------------------------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------|
| <b>591</b><br>OF 1000 VAULTS LEFT OPEN<br>AND UNDERWATER AT -95% | <b>57%</b><br>OF TOTAL DEBT LATENT AT<br>P95 — UNCOVERED, STILL<br>CIRCULATING | <b>61%</b><br>DRAWDOWN PAST WHICH NO<br>AUCTION FILLS AT ANY<br>PRICE | <b>0%</b><br>"REALISED" LOSS —<br>BECAUSE NOTHING IS<br>REALISED AT ALL |
|------------------------------------------------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------|

KFIAT realised losses *fall* as the crash deepens — 5% at -85%, 3% at -95%. That is not resilience. Losses fall because liquidations stop happening: liquidators run out of capital, the auction floor offers 15% while the market demands 21%, and nothing clears. The debt does not disappear. It becomes **latent**.

### WHY THIS MATTERS MORE HERE THAN ANYWHERE ELSE

For an immutable protocol this is *the* critical property: **there is nobody to unfreeze it**. No DAO to raise the MCR in an emergency, no admin key to inject capital. The frozen state persists until the market recovers on its own, or until someone forks.

### The fix, and what it costs

| CONFIGURATION                                     | LATENT P95 | ENRK LOSS P95 | KFIAT LOSS | CALM-MARKET COST |
|---------------------------------------------------|------------|---------------|------------|------------------|
| Floor 85% — as originally specified               | <b>46%</b> | 0%            | 5%         | baseline         |
| Floor 75%, no redemption                          | <b>0%</b>  | 11.2%         | 14%        | +2.4%            |
| <b>Floor 75% + redemption — the target design</b> | <b>0%</b>  | <b>5.0%</b>   | <b>2%</b>  | <b>+2.4%</b>     |

At KAS -85%. Moving the floor eliminates the freeze outright; adding redemption then halves the realised cost of doing so. The two are complementary because they act at different moments — redemption is **preventive**, draining weak vaults while they are still above 100% ICR; the deeper floor is **curative**, clearing what redemption could not reach.

### CLEARING BEATS FREEZING

Frozen: 0% + 5% realised, but **46% of debt unbacked and circulating**. Clearing: 1% + 14% realised, and **no hole at all**. A system that unfreezes takes less total damage, because it avoids the reflexive spiral and sells while depth still exists.

### Secondary findings

#### The hedge is not free

Alt 3 (pure Kaspia) leaves a 10% latent hole at -85% where the deployed Alt 3.5 leaves 17%. The uncorrelated energy component costs ~7 points of solvency. Not a reason to change the formula — a reason not to claim the hedge is free.

#### Energy crisis scenario

2022 pattern, energy +60% during the crash: latent hole 17% → 19%. Real, but far milder than predicted verbally beforehand. The model corrected an intuition that was too pessimistic.

#### Distrust these two numbers

Every figure on this page is dominated by `impact_coefficient` (0.08, square-root) and `daily_liquidation_capacity` (5% of debt/day). They are **assumptions, not measurements**. Vary them and re-run before trusting anything here.

## 09 Three paths rejected with numbers

Each of these looked right, and each was dropped because the model disagreed. They are documented because a design is only credible if its rejected branches are visible.

### REJECTED

#### An uncorrelated gold reserve

The idea: hold a reserve in an asset uncorrelated with KAS, to cover shortfalls a crypto crash produces.

**Why it fails:** a finite, publicly visible reserve that pays whoever arrives first creates a first-mover advantage — the rational response to stress becomes exiting before it empties. That is a bank-run trigger, and it survives the two genuine improvements a modern version has (no custodian, no human discretion), because the flaw comes from *finiteness and visibility*, not from custody. The only sound form pays no individual and is consumed pro rata across all affected holders — **and pro-rata socialisation requires global aggregate state, which Kaspa L1's UTXO model cannot express**. Separately, the 250% ICR result showed the same protection was available with no trusted party at all.

### REJECTED

#### Liquidity-style Recovery Mode

The idea: when system-wide collateralisation falls below a threshold, automatically raise the MCR so more vaults become liquidatable. Governance-free, proven elsewhere.

**Why it fails:** modelled at -85% it triggered on **47 of 60 days** and improved the p95 latent hole by **zero points**. Recovery Mode makes more vaults *eligible*; the failure is in *execution*. Adding vaults to a queue that is not moving changes nothing. In Liquidity it works because a Stability Pool absorbs liquidations without needing an auction bidder — the two are a pair, and ENRK's pool cannot buy.

**Consequence:** this corrected an earlier claim that losing Recovery Mode was "a genuine loss" and the most significant cost of the L1 path. It is not. Its absence costs essentially nothing — which removed the only serious technical objection to Kaspa L1.

### REJECTED

#### Raising the ICR minimum to 250%

The idea: more collateral per unit of debt buys more safety.

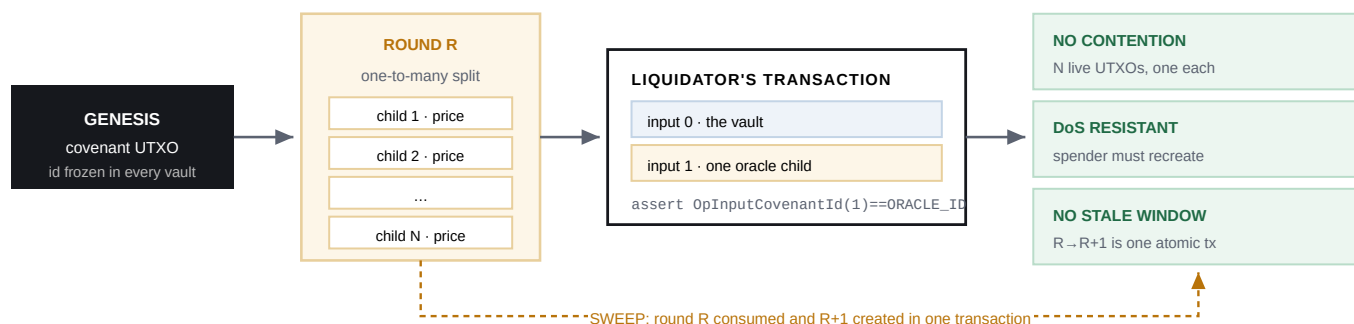
**Why it fails:** once the auction floor is corrected, 200% → 250% moves ENRK p95 loss from 11.2% to **11.6%** — **slightly worse** — while costing users 25% capital efficiency. A higher ICR makes vaults survive longer and therefore **liquidate later, at worse prices**. Liquidating early beats holding more collateral. Raising the MCR fails for the same reason: MCR 200% moves the p95 hole from 46% to 44% and costs +5.2% in calm markets. Wrong lever.

### THE PATTERN IN ALL THREE

Every rejected mechanism tried to add *more* — more reserve, more eligibility, more collateral. The one that worked removed a constraint: let the auction descend far enough that someone actually bids. **An immutable protocol should prefer the fix that is a constant over the fix that is a mechanism**, because a constant cannot fail in a way nobody predicted.

## 10 The oracle, and the question everything waits on

Two of the four peg indices are native to Kaspa and need no oracle. The other two do. The construction below uses only shipped opcodes — and it has not been validated by anyone who wrote them.



Authenticity comes from covenant lineage, not a signature — Kaspa script has no opcode to verify a signature over an arbitrary message.

**Figure 7.** The covenant-lineage price feed. Under KIP-9 storage mass rules a rising price is free and a falling price costs mass proportional to  $N$  — which pushes the design toward frequent rounds of small amplitude, what a good oracle should do anyway.

### THE BLOCKER — FILED AS KASPA.NET/KIPS ISSUE #46, SEPTEMBER 2026

1. Can a covenant reliably assert `OpInputCovenantId` on a *sibling* input against an embedded constant? 2. Does a one-to-many split preserve `covenant_id` across all authorised children? 3. What is the compute-mass ceiling on input count when every input runs a covenant script? 4. What are SilverScript's script-size limits? 5. Is a several-hundred-input sweep transaction reliably block-includable?

If the answer to (1) or (2) is no, the L1 path closes and this design deploys elsewhere or waits. Better to learn it from a core developer now than from an auditor in six months.

#### What is settled regardless of layer

- Fallback behaviour is hardcoded — no vote on which oracle to trust.
- No multi-sig can override the circuit breaker.
- If the feed degrades, the system **pauses** rather than accepting a suspect price.
- Hashrate and fees are chain-native: two of four indices need no oracle at all.

#### Why the Igra EVM path was set aside

Solidity, Liquity's proven lineage, a deep audit market — on engineering grounds it is the safer route. It was set aside because collateral would sit with a custodian: both documented KAS bridges are operator-controlled, one a 3-of-5 threshold whose five signers are run by a single organisation and are not publicly named. Both are capped near **200,000 KAS per 24 hours** — roughly 500 days to unwind a 100M KAS protocol.

**An immutable protocol on a custodial bridge is immutable in the part that matters least.**

## 11 Status, risks, and what is not claimed

### Where the work stands

|                                 |             |
|---------------------------------|-------------|
| Economic design                 | COMPLETE    |
| Peg formula, backtested         | COMPLETE    |
| Full-protocol crash stress test | PUBLISHED   |
| Frozen parameter specification  | COMPLETE    |
| Execution target assessment     | COMPLETE    |
| Covenant oracle construction    | UNVALIDATED |
| Redemption (mechanism #1)       | NOT BUILT   |
| Implementation                  | NOT STARTED |
| Audit                           | NOT STARTED |

### The four risks of immutability

1. **The peg weights are genuinely bad** → the system destabilises; a fork costs users a migration but preserves decentralisation.
2. **The oracle sources degrade** → the circuit breaker pauses the system; temporary freeze, not permanent failure.
3. **Collateral ratios are too strict** → vaults are uneconomic to open; a fork with different ratios competes.
4. **A critical bug reaches mainnet** → it cannot be patched, ever. **This is the real cost, and no testnet fully removes it.**

#### AND THE SPECIFIC RISK THIS PROJECT CARRIES

The protocol has been shown to **freeze** rather than explode under a deep crash, and there is nobody to unfreeze it. The 75% floor and redemption are what address that — **and redemption is not yet built**. Shipping without it would be shipping a known defect.

### What is not claimed

- No code is deployed. No audit has been performed.
- The Rust implementation (~5,000 lines, 119 passing tests) targeted an EVM rollout and **does not deploy**. It survives as an executable specification and a differential-testing oracle, nothing more.
- The oracle construction is derived from KIP text and has **not been validated by anyone who wrote those KIPs**.

- Three sizing questions are unresolved: compute-mass ceiling on input count, SilverScript size limits, block inclusion for a large sweep.
- Being first on these primitives is **a risk, not a feature**. Immutable code plus three-month-old primitives plus no auditors with covenant experience is a combination taken seriously here.
- Two model parameters dominate every figure and are assumptions, not measurements.

### What the ecosystem gets

#### A flagship for Toccata

KIP-10, 17 and 20 exercised adversarially in production with real value at stake.

#### Free adversarial QA

Everything found gets published — including what is found against us, as this document shows.

#### The first covenant audit methodology

Nobody has audited a SilverScript covenant system for immutable deployment. Whoever does it first produces the tooling and the pitfall catalogue every later Kaspas DeFi project inherits.

#### Proof that L1 covenants carry real DeFi

Kaskad chose Igra. KRON is on L1 but is an AMM, not a credit system. A working CDP stablecoin would be the strongest available demonstration.

#### THE ONE GOVERNANCE MOMENT

Before deployment: code review, independent audit of both code and economics, adversarial scenario testing, intensive scrutiny. Then a single decision — deploy as-is accepting the frozen rules, or do not deploy, or wait and demand more testing. **After deployment: none. Use it, or fork it.**

An economic audit matters more here than a second code audit. The freeze found in the stress test is a design defect in *perfectly correct code* — no code auditor would have caught it. That is how stablecoins die.